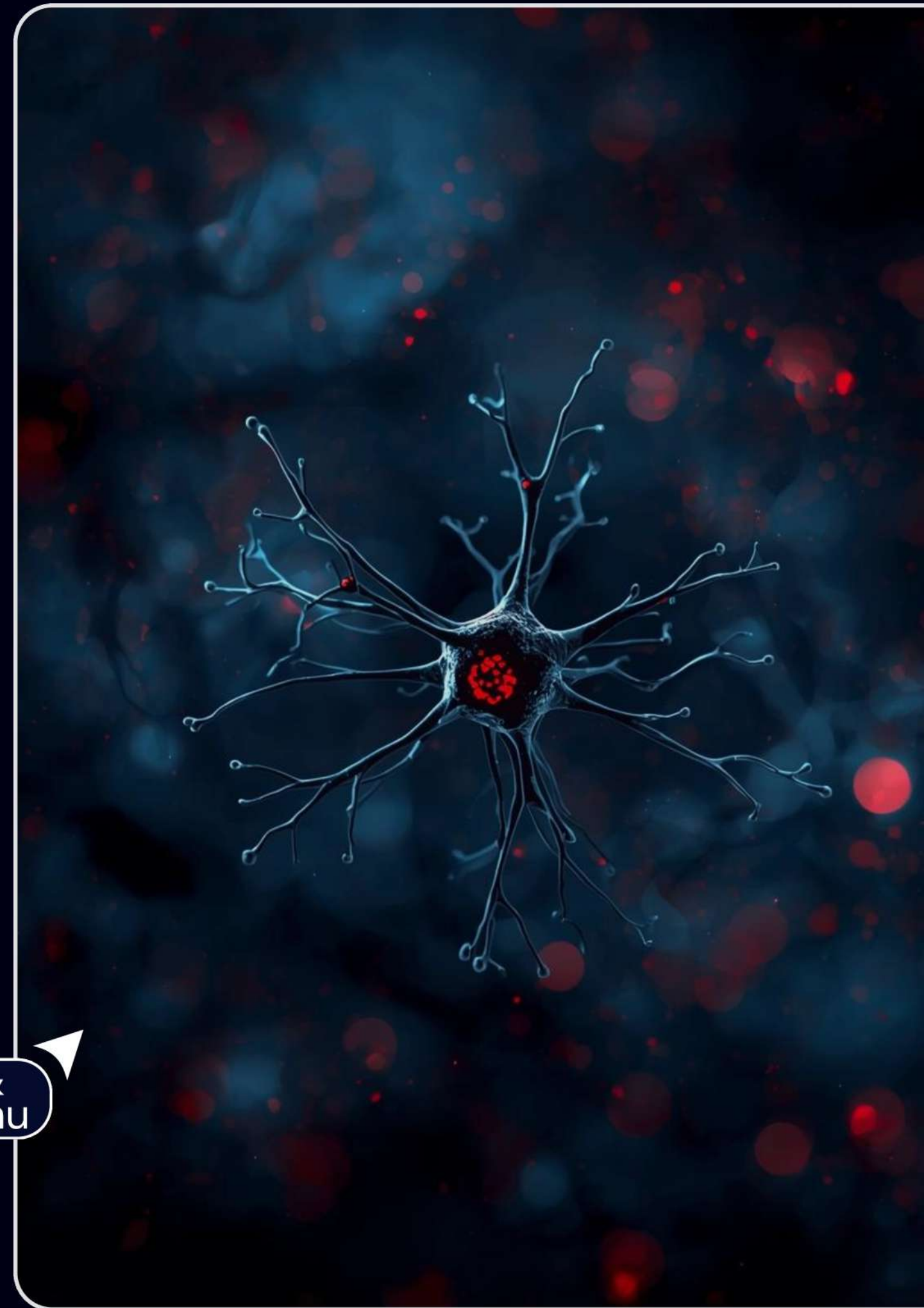


Vaani / वाणी / વાણી

Low Cost Edge AI Device for Screening of Neurodegenerative Diseases

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Understanding ALS

What is ALS?

- Neurodegenerative disease
- Affects motor neurons
- Causes loss of speech, movement, swallowing

Symptoms

- Muscle twitching
- Weakness in limbs
- Slurred / unstable speech
- Voice fatigue
- Paralysis

Importance For Early Detection

- ALS detected late (80% neuron loss)
- Early detection improves outcomes
- Speech changes appear BEFORE major symptoms
- Speech = earliest biomarker

Amyotrophic lateral sclerosis

Why Diagnosis Is Late

- Symptoms appear only after 80% neuron damage
- Neurodegeneration begins silently

Problems with Existing Diagnostic Methods

- Traditional Tools: MRI, PET Scan, Neurological Exams
- Very expensive
- Require hospital-grade equipment
- Time-consuming
- Not suitable for rural or low-resource settings

Late Diagnosis of ALS & Neurodegenerative Diseases

Lack of Affordable Early Screening

- No simple device to test speech biomarkers
- Zero accessibility for rural patients
- Clinical tests cost ₹ 10,000+
- No household-level screening available

Impact of Late Diagnosis

- ALS fatal within 2–5 years
- Treatment success drops drastically
- Families face huge financial burden
- Approx: 331,000 people living with ALS worldwide
- Around 58,000 new ALS cases per year
- ALS incidence rising steadily
- Mortality increasing each decade

The Gap

- No low-cost, portable, AI-powered, voice-based device

No low cost, portable solution & No AI powered, voice based device

VAANI: Edge AI Device for Screening of Neurodegenerative Diseases

Revolutionizing ALS diagnostics

VAANI is designed for **earlier, more accurate** detection of ALS. By leveraging cutting edge AI technology, our system analyzes patient data to identify symptoms at their earliest stages, providing **timely interventions** and improving patient outcomes significantly.



VAANI Process Flow

Data Acquisition

We collected data from pre existing, clinically recorded patient datasets rather than real time sensors. These datasets include verified speech recordings of ALS and Parkinson's patients, ensuring accurate and medically relevant training data.



AI Analysis

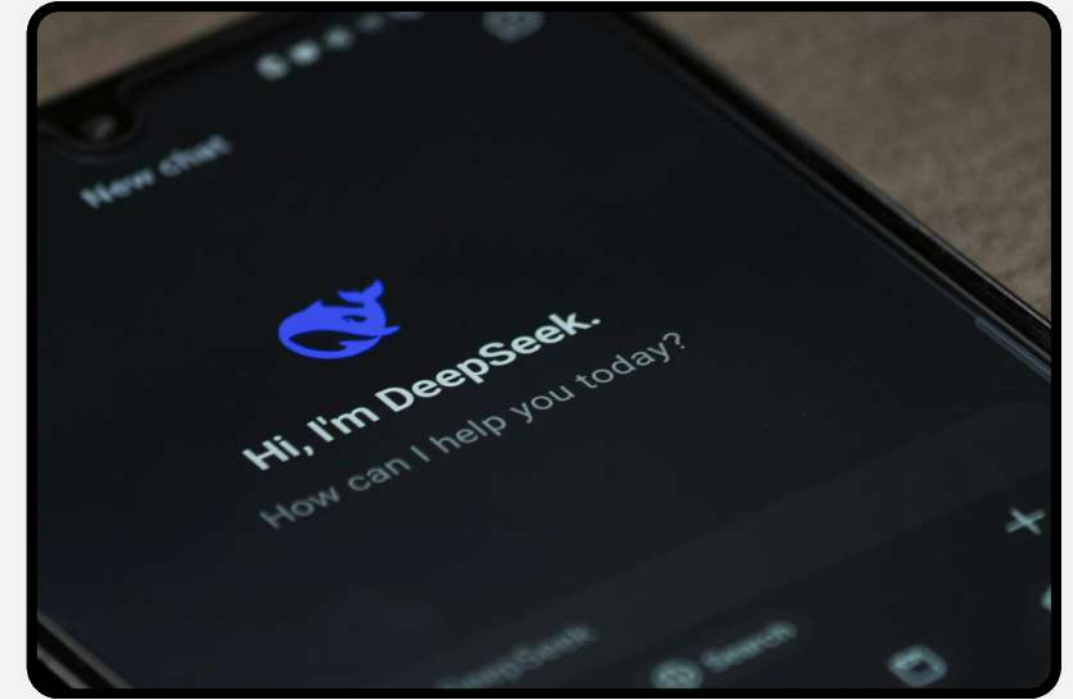
Based on the collected clinical speech data, the AI model analyzes sudden voice spikes, irregular pauses, unstable pitch, and action word distortion. These abnormalities often appear in the early stages of ALS, especially due to weakening of motor neurons controlling the vocal cords.

```
resp_iter = self.stub.get_response_iter()
statuses = {}
async for data in resp_iter:
    status = Status(
        status_id=data.id, name=data.name
    )
    statuses[status.name] = status
return statuses
```

Innovation and Technology

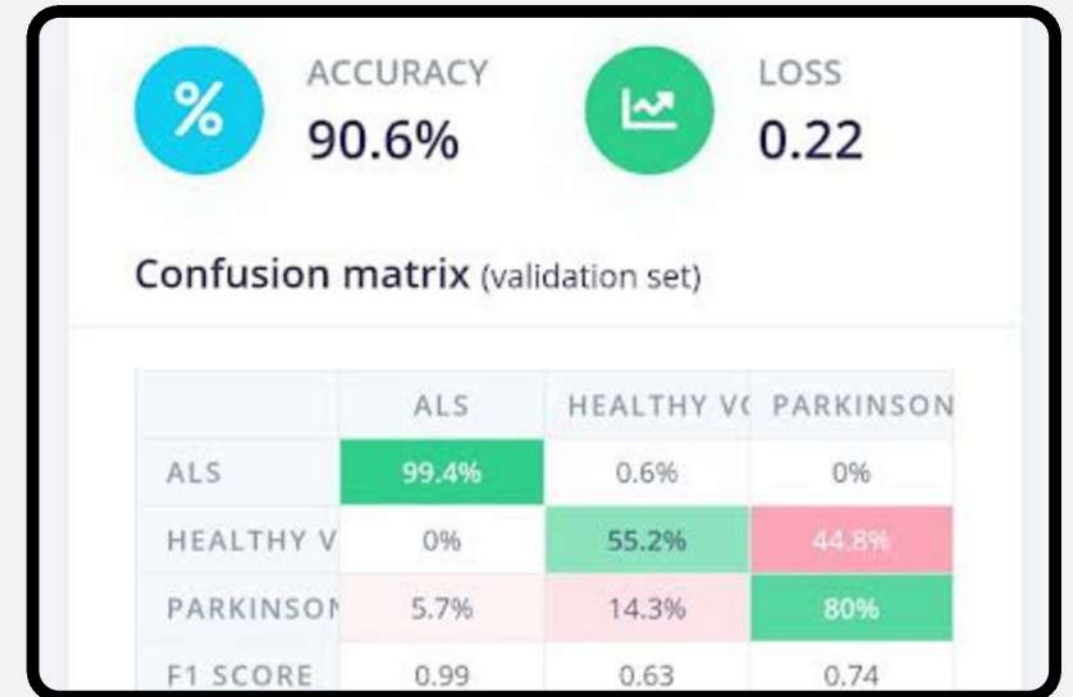
AI Driven

We use AI to detect tiny speech changes, like pitch instability and voice tremors, that appear early in ALS. A lightweight neural network analyzes patient audio data and runs directly on the ESP32, enabling fast, offline, low cost ALS screening.



Machine Learning

MFCC based speech degeneration classification model powered by Edge AI, utilizing advanced algorithms to detect early ALS related vocal changes with high accuracy.



Key Features of VAANI

Early Detection

VAANI identifies **early signs** of ALS, enabling timely intervention and better patient management.

Non Invasive

Our approach requires **no invasive procedures**, making it safer and more comfortable for patients.

High Accuracy

Utilizing advanced algorithms, VAANI ensures **high diagnostic accuracy** for ALS, minimizing false positives.

Personalized Medicine

VAANI tailors intervention strategies based on individual **patient profiles**, enhancing treatment effectiveness.

The Vaani Team

Parikshitsinh Jadeja

The backbone of building VAANI, designed and assembled the complete hardware, created the CAD models, and handled all video editing, documentary production, and presentation development.



Shubhranshu Jha

The driving force behind AI/ML development, engineered the full machine learning pipeline and served as the main speaker for our team.



THANK YOU

For further inquiries, please contact us at parikshitjadeja15@gmail.com